

SIH25009 · Smart India Hackathon



# Invertis EcoVerse

*Play. Learn. Sustain.*

[eco-verse-eight.vercel.app](https://eco-verse-eight.vercel.app)

Flutter App · Scan QR

## Team

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# Problem Statement

*Why do campus sustainability efforts fail to drive real, measurable behavioral change?*



## No Engagement

Eco-actions are passive campaigns — students don't participate consistently



## Actions Can't Be Verified

Checkbox missions are trivially fakeable — zero proof, zero trust



## No Institutional Data

Admins have no live dashboard for CO<sub>2</sub> offset or department sustainability



## No Retention Loop

No XP, streaks, or social dynamics — students disappear after day one

*EcoVerse answers all four — AI verification + Gamification + Admin analytics in one platform*

# Our Solution

*Dual-platform Green Campus OS — Web (React) + Mobile (Flutter) — one shared Firebase backend*



## AI Eco Lens

Photo proof → Gemini 2.0 Flash verifies  
→ XP awarded instantly



## Admin Dashboard

Live CO<sub>2</sub> offset, XP totals, dept.  
leaderboard & audit logs



## Campus Social Hub

Feed, Study Circles, join flows, circle chat  
& verified badges



## Swap Shop

Circular economy — list books/gear,  
claim items, earn XP



## QR Eco-Zones

Scan physical eco-spots, verify presence,  
duplicate-proof



## Gamification Engine

XP, levels, eco-streaks, daily challenges,  
dept. competition

**AI Verification + Social Impact + Gamification = Invertis EcoVerse**

# Key Features



## AI Eco Lens

Gemini 2.0 verifies proof photos



## XP Gamification

Levels, streaks & daily challenges



## Dept. Leaderboard

BTech vs MBA vs LLB vs BCA



## Social Hub

Circles, posts, chat & moderation



## QR Eco-Zones

Location-based XP check-ins



## Swap Shop

Circular campus marketplace

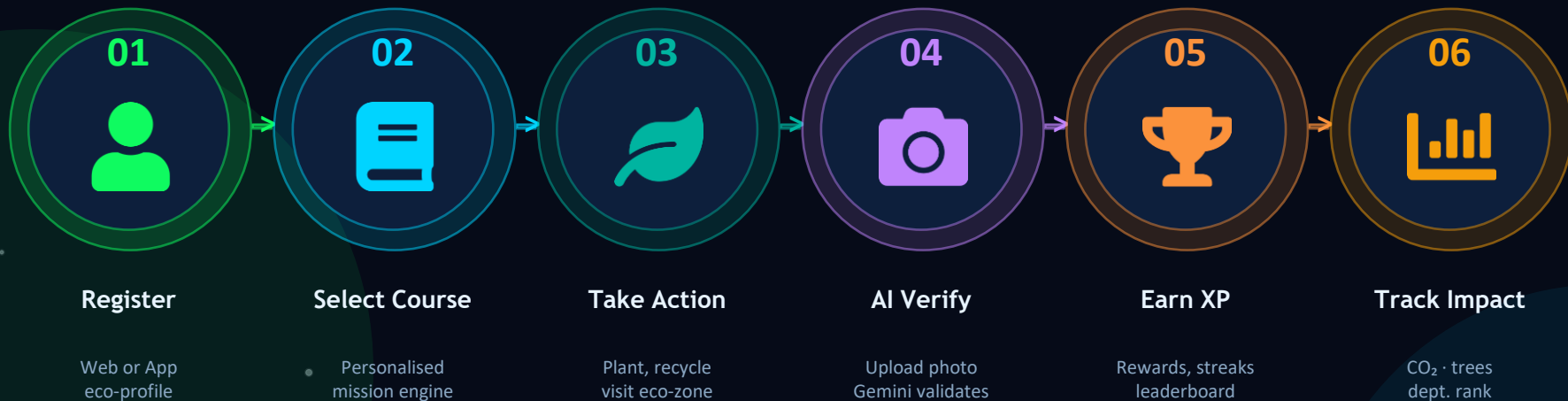


## Admin Dashboard

Analytics, moderation & audit logs

# How It Works

*From first login to verified eco-impact — 6 steps, two platforms, one mission*



Every verified action earns XP & creates measurable real-world eco-impact

# Technology Used

## WEB PLATFORM [eco-verse-eight.vercel.app](https://eco-verse-eight.vercel.app)

### Frontend

React 18 + Vite (SPA)

Framer Motion animations

React Router DOM

### Backend

FastAPI (Python)

Firebase Admin SDK

Gemini 2.0 integration

### Database

Firebase Realtime DB

Firebase Auth + Storage

Shared with mobile app

## MOBILE APP Flutter + Firebase (Scan QR to install)

### Frontend

Flutter (Dart)

Riverpod state mgmt

go\_router navigation

### Backend

Firebase Realtime DB

Firebase Auth  
(anon+email)

Firebase Storage

### AI + Tools

Google Gemini API

mobile\_scanner (QR)

pdf + printing (certs)

## SHARED LAYER

Gemini 2.0 Flash · Firebase Project ecoverse-51c09 · institutions/invertis\_001/ · users · scores · eco\_zones · audit\_logs

APK (Android) + PWA (Web) — Both deployment-ready for SIH demo

# Try It Live

*Both platforms are live and demo-ready right now*



## Web Platform

[eco-verse-eight.vercel.app](https://eco-verse-eight.vercel.app)

*Open in any browser — no install needed*



## Mobile App (Android)



*Scan to download & install APK*

# Impact & Future Scope

## IMPACT

- Promotes sustainability as a lifestyle, not a checkbox
- AI-verified actions ensure trust & real behavioral change
- Tracks CO<sub>2</sub> offset + trees planted per user & department

## FUTURE SCOPE

- AR-based immersive eco verification
- Inter-college & national competitions
- Government / NGO / NAAC integration
- Open-source multi-institution rollout

**10K+**

Target Users  
(Year 1)

**50+**

Eco Courses

**25K**

Tonnes CO<sub>2</sub>  
Tracked

**100+**

Colleges  
Reachable

**Building a Greener Future through Technology**